Vidya Pratishthan's
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Vidyanagari, Baramati, Pune - 413133

# Department of Artificial Intelligence and Data Science

**Academic Year**:2025-26

## INDEX

**Class:**

| Sr. No. | Name of the Assignment | Conducted Date | Checked Date | Page No. | Sign | Remark |
|---|---|---|---|---|---|---|
| **1** | Set up a Git repository for a data science project and push it to GitHub to manage version control, track changes, and ensure code stability. | | | | | |
| **2** | Create feature branches for the project, merge them, and resolve conflicts to maintain separate workspaces for team collaboration. | | | | | |
| **3** | Fork an open-source AI project, create a feature branch, submit a pull request, and address conflicts during the review process. | | | | | |
| **4** | Research and explain how DevOps principles, like CI/CD, can improve deployment and updates for machine learning projects. | | | | | |
| **5** | Set up Jenkins to trigger automated builds whenever new code is pushed to GitHub for your AI/DS project. | | | | | |
| **6** | Create a Docker container for your AI model and test it on your local machine to ensure consistent performance across environments. | | | | | |
| **7** | Use Docker Compose to manage multiple containers for a web-based AI app, including containers for the AI model, database, and frontend. | | | | | |
| **8** | Set up a local Kubernetes cluster with Minikube to deploy and scale your Dockerized AI application. | | | | | |
| **9** | Implement auto-scaling for your AI app on Kubernetes and monitor its health and performance metrics. | | | | | |

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| **10** | Set up a CI/CD pipeline with Jenkins, Docker, and Kubernetes to automate the development-to deployment process for your AI model. | | | | | |

## CERTIFICATE

This is to certify that Mr./Miss _____
Roll No. _____has Satisfactorily Completed the practical work of the subject **VSEC - DevOps and Git** for the First /Second Semester of 2025- 26.

**Date:** **Staff In-Charge** **H.O.D.** **Principal**